

Theo:-

Let $\langle G, * \rangle$ be a finite cyclic gp generated by $a \in G$. If G is of order n , i.e. $|G| = n$, then $a^n = e$, so that $G = \{a, a^2, a^3, \dots, a^n = e\}$

Furthermore, n is the least positive integer for which $a^n = e$.

Pf:

Suppose assume for some +ve integer $m < n$, $a^m = e$. Since G is cyclic gp any elt of G can be written as a^k for some $k \in \mathbb{I}$. From Euclid's algorithm

$$k = mq + r, \text{ where } q \in \mathbb{I} \text{ and } 0 \leq r < m$$

$$\Rightarrow a^k = a^{mq+r} = (a^m)^q * a^r = a^r$$

i.e. every elt of G can be expressed as a^r for some $0 \leq r < m$, thus implying G has atmost m distinct elts.

$$\text{i.e. } |G| = m < n \Rightarrow \Leftarrow$$

Hence, $a^m = e$ for $m < n$ is not possible.

We show $a, a^2, a^3, \dots, a^n$ are all distinct

where $a^n = e$

$$\text{Suppose } a^i = a^j \text{ for } i < j \leq n$$

$$\Rightarrow a^{j-i} = e \text{ where } j-i < n$$

Which is again $a \Rightarrow \Leftarrow$

Subgroups and Homomorphisms.

Defn: Let $\langle G, * \rangle$ be a group and $S \subseteq G$ be such that it satisfies

- i) $e \in S$, where e is the identity of $\langle G, * \rangle$
- ii) for any $a \in S$, $a^{-1} \in S$
- iii) for $a, b \in S$, $a * b \in S$

Then $\langle S, * \rangle$ is called a subgroup of $\langle G, * \rangle$

For any group $\langle G, * \rangle$, naturally $\langle \{e\}, * \rangle$ and $\langle G, * \rangle$ are trivial subgroups of $\langle G, * \rangle$. All other subgroups of $\langle G, * \rangle$ are called proper subgroups.

Let $\langle G, * \rangle$ be a group and $a \in G$ then $a^r \in G$ for $r \in \mathbb{I}$. This means that cyclic group generated by " a " must be a subgroup of $\langle G, * \rangle$

Consider $\langle \mathbb{Z}_6, +_6 \rangle$

- i) $[1]$ and $[5]$ generate $\langle \mathbb{Z}_6, + \rangle$
- ii) $[2]$ and $[4]$ generate the subgroup $\langle \{[0], [2], [4], +_4 \rangle$
- iii) $[3]$ generate $\langle \{[0], [3]\}, +_4 \rangle$

Theo: A subset $S \neq \emptyset$ of G is a subgroup of $\langle G, * \rangle$ iff for any pair of elts $a, b \in S$, $a * b^{-1} \in S$.

Assume S is subgroup, then if $a, b \in S$ then $b^{-1} \in S$ and $a * b^{-1} \in S$

Conversely assume if $a, b \in S$ then $a * b^{-1} \in S$.

tp S is subgroup.

- i) Let $a \in S$, by assumption $e = a * a^{-1} \in S$
- ii) now let $e, a \in S$ we have $a^{-1} = e * a^{-1} \in S$
- iii) for $a, b \in S$ by ii) $a^{-1}, b^{-1} \in S$
then $a * b = a * (b^{-1})^{-1} \in S$.

Hence $\langle S, * \rangle$ is sub gp of $\langle G, * \rangle$